## Experiment-8

## Support Vector Machines

**Aim:**

To implement Support Vector Machine classifiers using linear and non-linear kernels for classification tasks, and to evaluate their performance on linearly and non-linearly separable datasets.

**Theory:**

Support Vector Machines (SVMs) were formally established as a statistical learning framework grounded in the principle of Structural Risk Minimization (SRM), by Cortes and Vapnik (1995). The fundamental objective of SVM resides in determining an optimal decision boundary that maximizes the separation margin between different classes in a given feature space.

The goal of the Support Vector Machine is to find the hyperplane that maximizes the margin between two classes. This principle ensures improved generalization performance by minimizing the upper bound on the generalization error rather than merely minimizing empirical risk.

**I. Hyperplane: -**

A hyperplane is **a flat affine subspace of one dimension less than its ambient space**. In an n-dimensional space, a hyperplane has (n-1) dimensions. For a linearly separable dataset, the SVM seeks a separating hyperplane defined by:

$$w^{\mathrm{T}} x + b = 0$$

where $\boldsymbol{w}$ is the weight vector which is normal to the hyperplane and $\boldsymbol{b}$ is the bias term.
**Note -** An optimal hyperplane is the one that maximizes the margin between the classes.

**II. Support Vectors: -**

A margin is defined as the perpendicular distance between the hyperplane and the nearest data points from each class. These points are known as support vectors.

**III. Margin Maximization: -**

The margin maximization problem can be formulated as:

$$\min_{w,b} \frac{1}{2} \| w \|^2$$

with constraint,

$$y_i(w \cdot x_i + b) \geq 1$$

for all training samples $(x_i, y_i)$. This formulation ensures that the separating hyperplane lies as far as possible from the closest data points, thereby enhancing the classifier's ability to generalize to unseen data.

IV. **Kernels: -** SVM employs the kernel trick to handle decision boundaries. The input data is implicitly mapped to a higher-dimensional feature space where separation becomes feasible. This transformation is performed through kernel functions without explicitly computing the coordinates in the high-dimensional space.

Common Kernel functions include:

- **Linear Kernel**: The linear kernel is suitable when the data is approximately linearly separable.

$$K(x_i, x_j) = x_i \cdot x_j$$

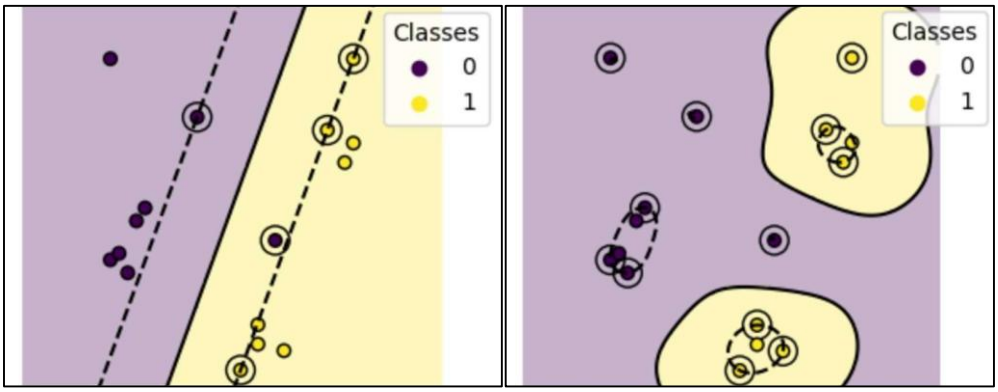
- **Radial Basis Function (RBF) Kernel**: The RBF kernel maps data into an infinite-dimensional feature space and is highly effective for capturing complex, non-linear relationships.

$$K(x_i, x_j) = e^{\frac{-|x_i - x_j|^2}{2\sigma^2}}$$

The variance ($\sigma^2$) controls the influence of individual training samples. Higher values lead to tighter decision boundaries.

The **RBF kernel** is particularly effective and is commonly used for modelling complex, non-linear relationships due to its radially localised response characteristics.

In the figure given below, Linear Kernel (left) is unable to separate a single 'Class 0' from 'Class 1' sample which RBF Kernel (right) is able to separate well. Mainly due to, RBF can get more meaningful support vectors (circled points) to classify sample points with help of better hyperplane formation.




**Merits of Support Vector Machines**

- **Good generalization performance:**
  SVMs focus on maximizing the margin between classes, which helps the model perform well on unseen data and reduces overfitting, especially in high-dimensional datasets.
- **Works well for both linear and non-linear data:**
  By using different kernel functions such as Linear and RBF, SVMs can handle simple linearly separable data as well as complex non-linear patterns effectively.

- **Uses only important data points:**
  The model depends mainly on support vectors, which are the most critical data points near the decision boundary. This makes the classifier efficient and robust.

**Demerits of Support Vector Machines**

- **High training time for large datasets:**
  SVM training can be slow and computationally expensive when the dataset is very large, particularly when non-linear kernels are used.
- **Sensitive to parameter selection:**
  The performance of SVM strongly depends on choosing the right kernel and hyper-parameters. Incorrect values can lead to poor classification results.
- **Harder to interpret results:**
  Unlike simpler models such as Linear Regression or Decision Trees, especially SVMs with non-linear kernels do not provide clear insights into how individual features might affect predictions.

**Algorithm**

**Step 1:** Given training data with two classes (+1 and -1)

**Step 2:** Find hyperplane: $w \cdot x + b = 0$

- w = weight vector (perpendicular to hyperplane)
- b = bias term

**Step 3:** Define margin constraints:

- For class +1 points: $w \cdot x_i + b \geq +1$
- For class -1 points: $w \cdot x_i + b \leq -1$
- Combined: $y_i(w \cdot x_i + b) \geq 1$

**Step 4:** Margin width = $2/||w||$

**Step 5:** Optimization problem:

- Minimize: $(1/2)||w||^2$
- Subject to: $y_i(w \cdot x_i + b) \geq 1$ for all i

**Step 6:** Solve using Lagrange multipliers:

- Convert to dual form
- Find $\alpha_i$ values for each training point
- Support vectors are points where $\alpha_i > 0$

**Step 7:** For non-linear data, apply Kernel Trick:

- RBF Kernel: $K(x,x') = \exp(-\gamma||x-x'||^2)$
- Polynomial Kernel: $K(x,x') = (x \cdot x' + c)^d$
- Maps data to higher dimension where linear separation is possible

**Step 8:** For prediction:

- Calculate: $f(x) = \Sigma(\alpha_i \times y_i \times K(x_i,x)) + b$

If $f(x) \geq 0$ then it belongs to Class +1, otherwise, it belongs to class -1

# SVM - Practical Implementation

*Virtual Labs - Dayalbagh Educational Institute*

## Step 1: Importing Libraries

**# Importing Libraries**

```
# Numerical and data handling libraries
import numpy as np
import pandas as pd

# Visualization libraries
import matplotlib.pyplot as plt
import seaborn as sns
from mlxtend.plotting import plot_decision_regions

print("Loaded libraries")
```

**Output:**

```
Loaded libraries
```

**# Dataset utilities**

```
from sklearn.datasets import load_wine, make_moons
# Preprocessing and model utilities
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
# Models
from sklearn.svm import SVC
# Evaluation metrics
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    classification_report,
    confusion_matrix
)

print("Loaded Sklearn libraries")
```

**Output:**

```
Loaded Sklearn libraries
```

## Step 2: Loading Dataset

**# Loading and Reading Data**

```
wine = load_wine()
wine_df = pd.DataFrame(wine.data, columns=wine.feature_names)

wine_df["target"] = wine.target
wine_df
```

Output:

| | alco. | malic_a. | as | alcalinity_of. | magnes. | total_phe. | flavano. | nonflavanoid_phe. | proanthocya. | color_inten. | hu | od280/od315_of_diluted_.. | prol. | tar. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 14.23 | 1.71 | 2.4 | 15.6 | 127.0 | 2.80 | 3.06 | 0.28 | 2.29 | 5.64 | 1.0 | 3.92 | 1065. | 0 |
| 1 | 13.20 | 1.78 | 2.1 | 11.2 | 100.0 | 2.65 | 2.76 | 0.26 | 1.28 | 4.38 | 1.0 | 3.40 | 1050. | 0 |
| 2 | 13.16 | 2.36 | 2.6 | 18.6 | 101.0 | 2.80 | 3.24 | 0.30 | 2.81 | 5.68 | 1.0 | 3.17 | 1185. | 0 |
| 3 | 14.37 | 1.95 | 2.5 | 16.8 | 113.0 | 3.85 | 3.49 | 0.24 | 2.18 | 7.80 | 0.8 | 3.45 | 1480. | 0 |
| 4 | 13.24 | 2.59 | 2.8 | 21.0 | 118.0 | 2.80 | 2.69 | 0.39 | 1.82 | 4.32 | 1.0 | 2.93 | 735.0 | 0 |
| .. | | | | | | | | | | | | | | |
| 173 | 13.71 | 5.65 | 2.4 | 20.5 | 95.0 | 1.68 | 0.61 | 0.52 | 1.06 | 7.70 | 0.6 | 1.74 | 740.0 | 2 |
| 174 | 13.40 | 3.91 | 2.4 | 23.0 | 102.0 | 1.80 | 0.75 | 0.43 | 1.41 | 7.30 | 0.7 | 1.56 | 750.0 | 2 |
| 175 | 13.27 | 4.28 | 2.2 | 20.0 | 120.0 | 1.59 | 0.69 | 0.43 | 1.35 | 10.20 | 0.5 | 1.56 | 835.0 | 2 |
| 176 | 13.17 | 2.59 | 2.3 | 20.0 | 120.0 | 1.65 | 0.68 | 0.53 | 1.46 | 9.30 | 0.6 | 1.62 | 840.0 | 2 |
| 177 | 14.13 | 4.10 | 2.7 | 24.5 | 96.0 | 2.05 | 0.76 | 0.56 | 1.35 | 9.20 | 0.6 | 1.60 | 560.0 | 2 |

Output:

```
178 rows x 14 columns
```

## # Display Wine dataset info

```
wine_df.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
#   Column                        Non-Null Count  Dtype  
---  ------                        --------------  -----  
0   alcohol                       178 non-null    float64
1   malic_acid                    178 non-null    float64
2   ash                           178 non-null    float64
3   alcalinity_of_ash             178 non-null    float64
4   magnesium                     178 non-null    float64
5   total_phenols                 178 non-null    float64
6   flavanoids                    178 non-null    float64
7   nonflavanoid_phenols          178 non-null    float64
8   proanthocyanins               178 non-null    float64
9   color_intensity               178 non-null    float64
10  hue                           178 non-null    float64
11  od280/od315_of_diluted_wines  178 non-null    float64
12  proline                       178 non-null    float64
13  target                        178 non-null    int32  
dtypes: float64(13), int32(1)
memory usage: 19.0 KB
```

## # Generate two-moons dataset

```
X_moons, y_moons = make_moons(n_samples=700, noise=0.15, random_state=42)
moons_df = pd.DataFrame(X_moons, columns=["Feature_1", "Feature_2"])
moons_df["target"] = y_moons
moons_df
```

Output:

| | Feature_1 | Feature_2 | target |
|---|---|---|---|
| 0 | -0.036777 | 1.017428 | 0 |
| 1 | 0.991181 | -0.542893 | 1 |
| 2 | 0.135605 | 0.095587 | 1 |
| 3 | 0.380490 | 0.882966 | 0 |
| 4 | -0.795374 | 0.193136 | 0 |
| ... | | | |
| 695 | 1.147087 | 0.520498 | 0 |
| 696 | 0.200663 | 0.598349 | 0 |
| 697 | -0.384952 | 0.547244 | 0 |
| 698 | 0.452570 | -0.174473 | 1 |
| 699 | 0.534527 | 0.910810 | 0 |

Output:

```
700 rows x 3 columns
```

## Step 3: Data Analysis

**# Data Analysis**

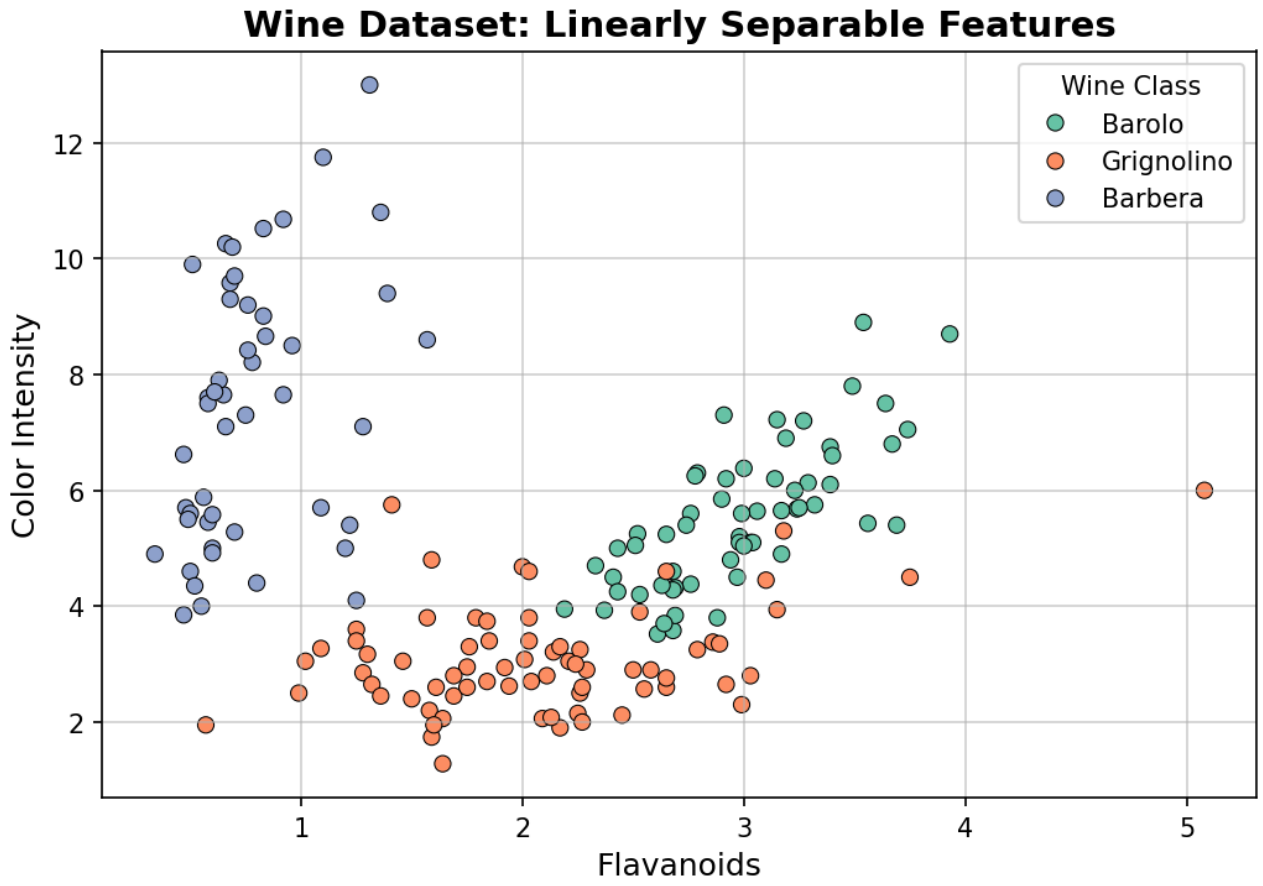
```
wine_df = pd.concat(
    [wine_df[["flavanoids", "color_intensity"]], wine_df["target"]],
    axis=1
)

wine_target_names = {
    0: "Barolo",
    1: "Grignolino",
    2: "Barbera"
}

wine_df["class_name"] = wine_df["target"].map(wine_target_names)
wine_df[['target', 'class_name']].value_counts()
```

Output:

```
target  class_name
1       Grignolino    71
0       Barolo        59
2       Barbera       48
Name: count, dtype: int64
```

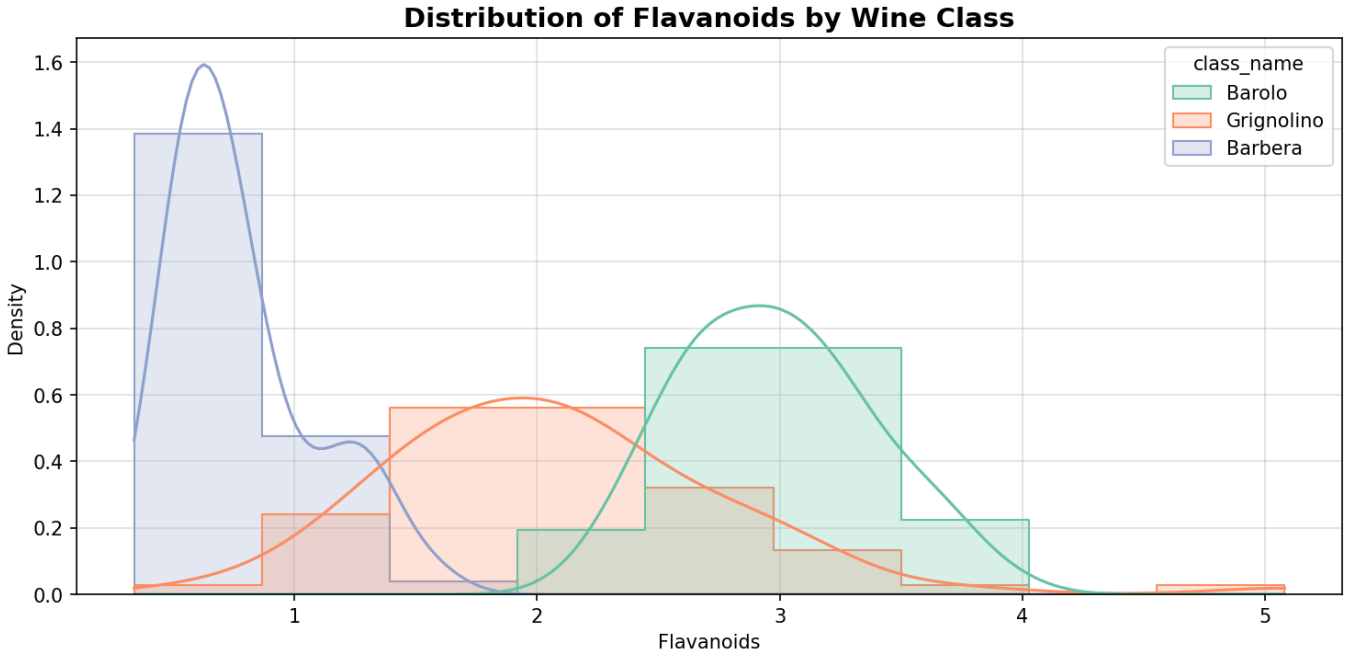
**# Visualizing Wine Dataset**

```
plt.figure(figsize=(7, 5))
sns.scatterplot(
    data=wine_df,
    x="flavanoids",
    y="color_intensity",
    hue="class_name",
    palette="Set2",
    s=40,
    edgecolor="black"
)
plt.title("Wine Dataset: Linearly Separable Features", fontsize=14, fontweight="bold")
plt.xlabel("Flavanoids", fontsize=12)
plt.ylabel("Color Intensity", fontsize=12)
plt.legend(title="Wine Class")
plt.grid(alpha=0.6)
plt.show()
```



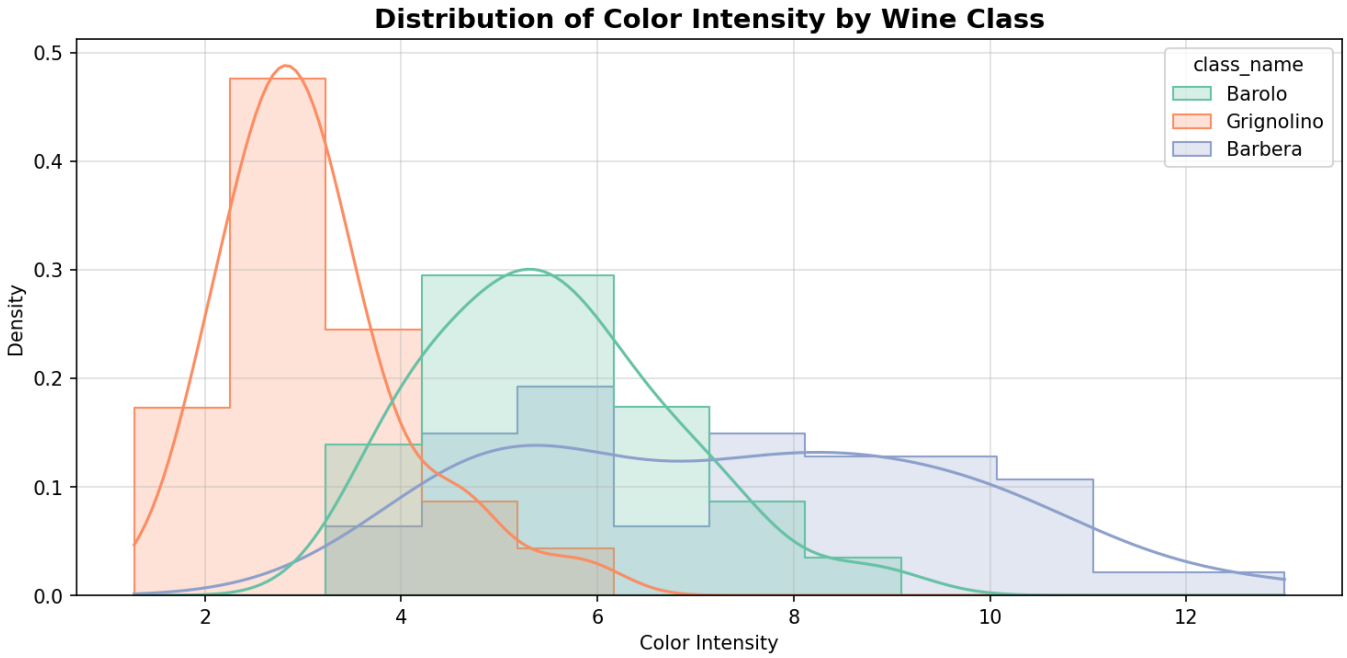

```
# Histogram of Flavanoids
```

```
plt.figure(figsize=(10, 5))
sns.histplot(
    data=wine_df,
    x="flavanoids",
    hue="class_name",
    kde=True,
    palette="Set2",
    element="step",
    stat="density",
    common_norm=False
)
plt.title("Distribution of Flavanoids by Wine Class", fontsize=14, fontweight="bold")
plt.xlabel("Flavanoids")
plt.ylabel("Density")
plt.grid(alpha=0.4)
plt.show()
```



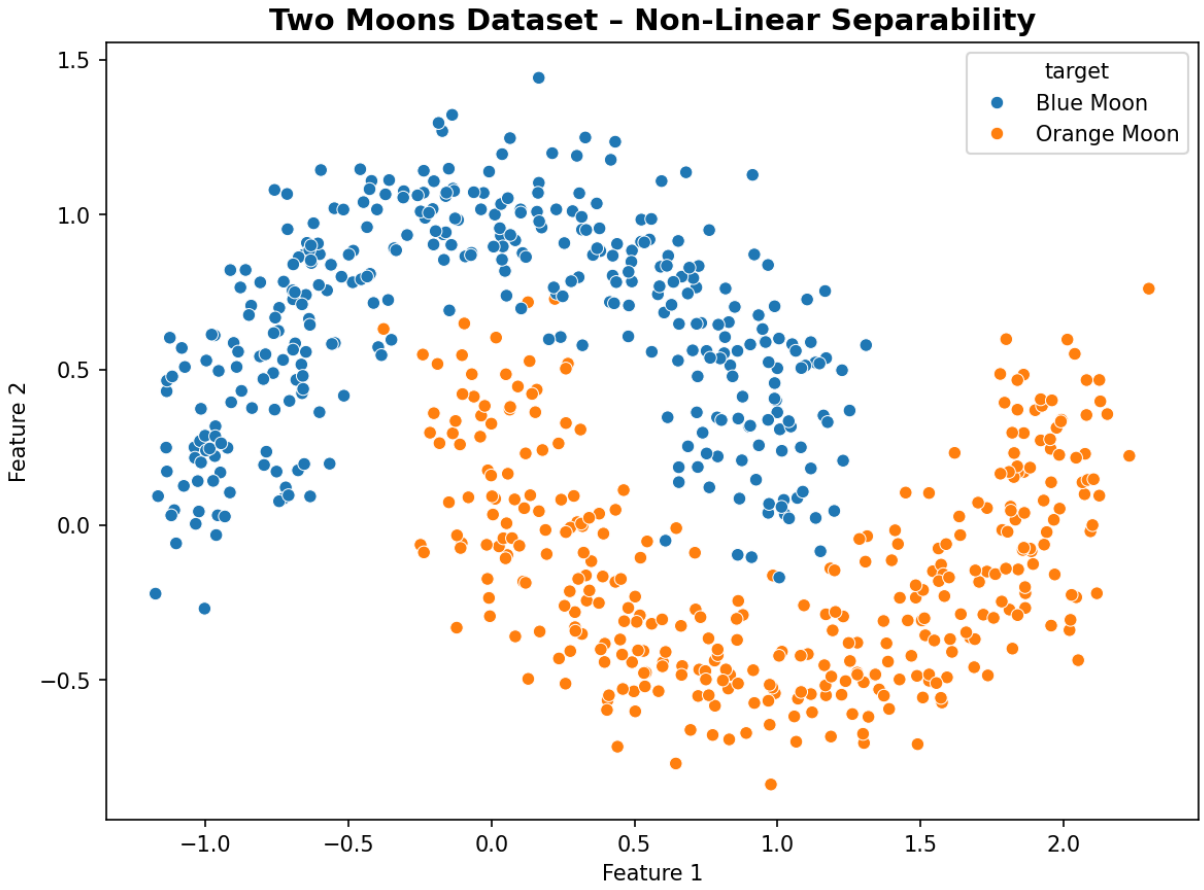

## # Histogram of Color Intensity

```
plt.figure(figsize=(10, 5))
sns.histplot(
    data=wine_df,
    x="color_intensity",
    hue="class_name",
    kde=True,
    palette="Set2",
    element="step",
    stat="density",
    common_norm=False
)
plt.title("Distribution of Color Intensity by Wine Class", fontsize=14, fontweight="bold")
plt.xlabel("Color Intensity")
plt.ylabel("Density")
plt.grid(alpha=0.4)
plt.show()
```




## # Visualizing Two Moons Dataset

```
plt.figure(figsize=(8,6))
sns.scatterplot(
    x=moons_df["Feature_1"],
    y=moons_df["Feature_2"],
    hue=moons_df["target"].map({0: "Blue Moon", 1: "Orange Moon"}),
    palette=["#1f77b4", "#ff7f0e"]
)
plt.title("Two Moons Dataset - Non-Linear Separability", fontsize=14, fontweight='bold')
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
```




## Step 4: Data Preprocessing

**# Data Preprocessing**

```
X_wine = wine_df[["flavanoids", "color_intensity"]]
y_wine = wine_df["target"]

Xw_train, Xw_test, yw_train, yw_test = train_test_split(
    X_wine, y_wine, test_size=0.2, random_state=42, stratify=y_wine
)
Xw_train.shape, Xw_test.shape, yw_train.shape, yw_test.shape
```

Output:

```
((142, 2), (36, 2), (142,), (36,))
```

**# Train/Test split for Two Moons dataset**

```
Xm_train, Xm_test, ym_train, ym_test = train_test_split(
    X_moons, y_moons, test_size=0.2, random_state=42, stratify=y_moons
)
Xm_train.shape, Xm_test.shape, ym_train.shape, ym_test.shape
```

Output:

```
((560, 2), (140, 2), (560,), (140,))
```

## # Feature Scaling

```
scaler = StandardScaler()
scaler
```

Output:

```
StandardScaler()
```

## # Scale Wine training and test data

```
Xw_train_scaled = scaler.fit_transform(Xw_train)
Xw_test_scaled = scaler.transform(Xw_test)

print("Wine data scaled successfully!")
```

Output:

```
Wine data scaled successfully!
```

## # View original Wine training data (before scaling)

```
Xw_train[:5]
```

Output:

| | flavanoids | color_intensity |
|---|---|---|
| 10 | 2.81 | 5.40 |
| 149 | 3.17 | 5.75 |
| 90 | 3.00 | 5.50 |
| 59 | 2.91 | 5.60 |
| 147 | 0.66 | 9.39 |

## # View scaled Wine training data (after scaling)

```
Xw_train_scaled[:5]
```

Output:

```
array([[ 0.73229212, -0.16746725],
[ 1.33318146,  0.30530313],
[ 1.006382  , -0.081509  ],
[ 0.81662747,  0.262324  ],
[-1.29175618,  1.47433535]])
```

## # Scale Two Moons training and test data

```
Xm_train_scaled = scaler.fit_transform(Xm_train)
Xm_test_scaled = scaler.transform(Xm_test)

print("Two Moons data scaled successfully!")
```

Output:

```
Two Moons data scaled successfully!
```

**# View original Two Moons training data (before scaling)**

```
Xm_train[:5]
```

Output:

```
array([[ 1.3723603 , -0.55081882],
[ 0.63613117,  0.78345978],
[-0.43531115,  0.96004336],
[ 0.41718644,  1.17721089],
[-1.13531808,  0.17220389]])
```

**# View scaled Two Moons training data (after scaling)**

```
Xm_train_scaled[:5]
```

Output:

```
array([[ 1.01466867, -1.57697415],
[ 0.17320777,  1.01862443],
[-1.05137949,  1.36213595],
[-0.07703149,  1.78459625],
[-1.85144083, -0.17046376]])
```

## Step 5: Model Training

**# Model Training**

```
# Traing linear Kernel SVM in wine dataset with 2 features
svm_linear_wine = SVC(kernel="linear")
svm_linear_wine.fit(Xw_train_scaled, yw_train)
svm_linear_wine
```

Output:

```
SVC(kernel='linear')
```

**# Training SVMs with linear and RBF kernels on two-moons toy dataset**

```
svm_linear_moon = SVC(kernel="linear")
svm_rbf_moon = SVC(kernel="rbf")

svm_linear_moon.fit(Xm_train_scaled, ym_train)
svm_rbf_moon.fit(Xm_train_scaled, ym_train)
print("Model trained")
```

Output:

```
Model trained
```

## Step 6: Model Evaluation

### # Model Evaluation

```
# Wine dataset case
wine_preds = svm_linear_wine.predict(Xw_test_scaled)
wine_preds
```

Output:

```
array([0, 2, 0, 0, 1, 0, 0, 0, 1, 2, 1, 2, 0, 1, 0, 1, 1, 0, 1, 1, 1, 1,
0, 0, 0, 1, 0, 2, 1, 2, 0, 2, 1, 2, 2, 2])
```

### # Wine dataset - Classification metrics

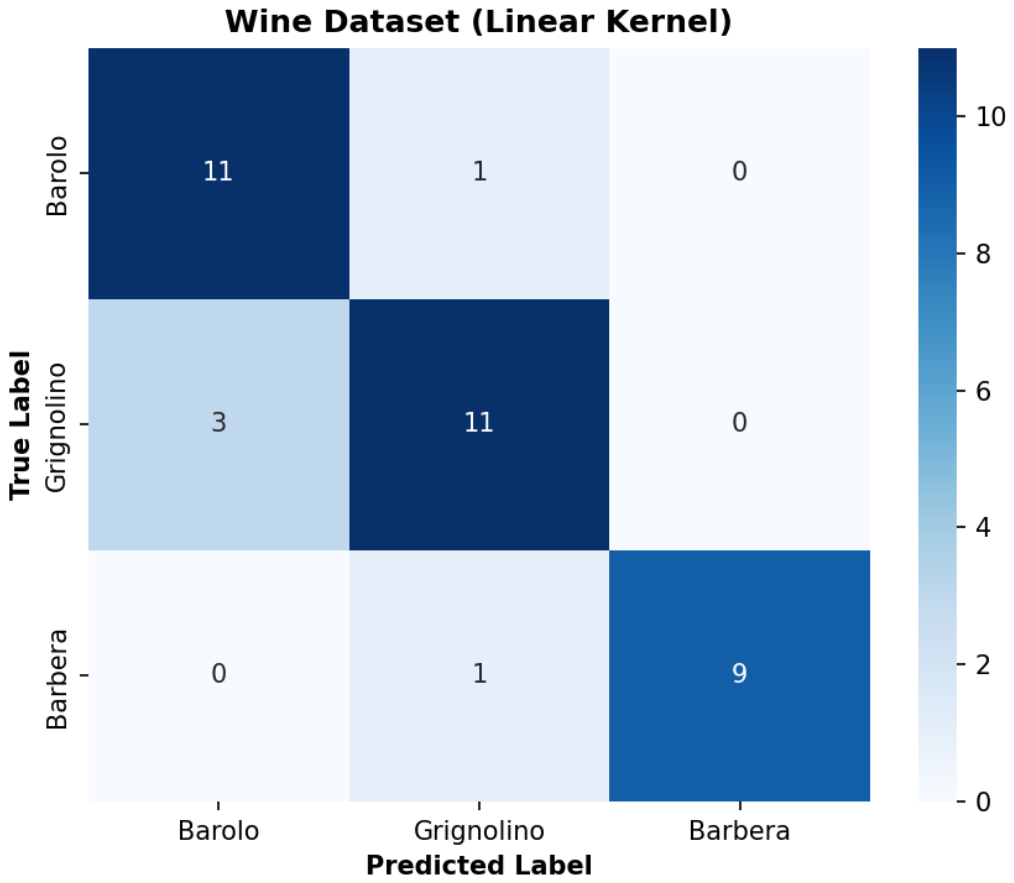
```
print("Accuracy:", accuracy_score(yw_test, wine_preds))
print("Precision:", precision_score(yw_test, wine_preds, average="macro"))
print("Recall:", recall_score(yw_test, wine_preds, average="macro"))
print("F1 Score:", f1_score(yw_test, wine_preds, average="macro"))

print("\\nClassification Report:\\n")
print(classification_report(yw_test, wine_preds))
```

Output:

```
Accuracy: 0.8611111111111112
Precision: 0.8772893772893773
Recall: 0.8674603174603175
F1 Score: 0.8694456940070975
Classification Report:
precision    recall  f1-score   support
0       0.79      0.92      0.85        12
1       0.85      0.79      0.81        14
2       1.00      0.90      0.95        10
accuracy                           0.86        36
macro avg       0.88      0.87      0.87        36
weighted avg       0.87      0.86      0.86        36
```

### # Confusion Matrix

```
labels = list(wine_target_names.keys())
class_names = list(wine_target_names.values())

plt.figure(figsize=(6, 5))
sns.heatmap(
    confusion_matrix(yw_test, wine_preds, labels=labels),
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=class_names,
    yticklabels=class_names
)

plt.title("Wine Dataset (Linear Kernel)", fontweight='bold')
plt.xlabel("Predicted Label", fontweight='bold')
plt.ylabel("True Label", fontweight='bold')
plt.tight_layout()
plt.show()
```




**# Two moon dataset case**

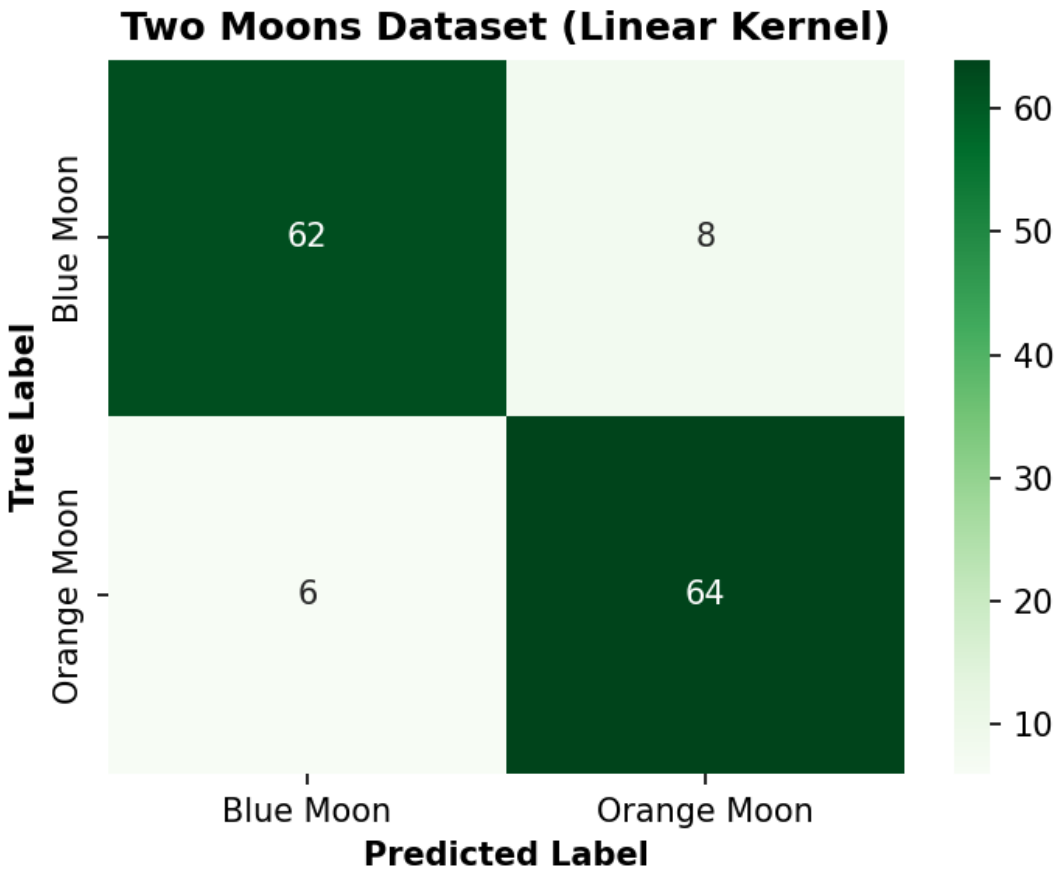
```
moon_linear_preds = svm_linear_moon.predict(Xm_test_scaled)
moon_linear_preds
```

Output:

```
array([0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0,
0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 1,
1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0, 1,
0, 1, 0, 1, 1, 0, 0, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0,
1, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0,
1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0,
0, 0, 1, 0, 0, 1, 1, 1])
```

**# Confusion Matrix**

```
class_names = ['Blue Moon', 'Orange Moon']

plt.figure(figsize=(5, 4))
sns.heatmap(
    confusion_matrix(ym_test, moon_linear_preds),
    annot=True,
    fmt="d",
    cmap="Greens",
    xticklabels=class_names,
    yticklabels=class_names
)

plt.title("Two Moons Dataset (Linear Kernel)", fontweight='bold')
plt.xlabel("Predicted Label", fontweight='bold')
plt.ylabel("True Label", fontweight='bold')
plt.tight_layout()
plt.show()
```




**# Two Moons - Linear kernel classification metrics**

```
print("Accuracy:", accuracy_score(ym_test, moon_linear_preds))
print("Precision:", precision_score(ym_test, moon_linear_preds, average="macro"))
print("Recall:", recall_score(ym_test, moon_linear_preds, average="macro"))
print("F1 Score:", f1_score(ym_test, moon_linear_preds, average="macro"))

print("\\nClassification Report:\\n")
print(classification_report(ym_test, moon_linear_preds))
```

Output:

```
Accuracy: 0.9
Precision: 0.9003267973856208
Recall: 0.8999999999999999
F1 Score: 0.8999795876709533
Classification Report:
precision    recall  f1-score   support
0       0.91      0.89      0.90        70
1       0.89      0.91      0.90        70
accuracy                           0.90       140
macro avg       0.90      0.90      0.90       140
weighted avg       0.90      0.90      0.90       140
```

## # Two moon dataset case

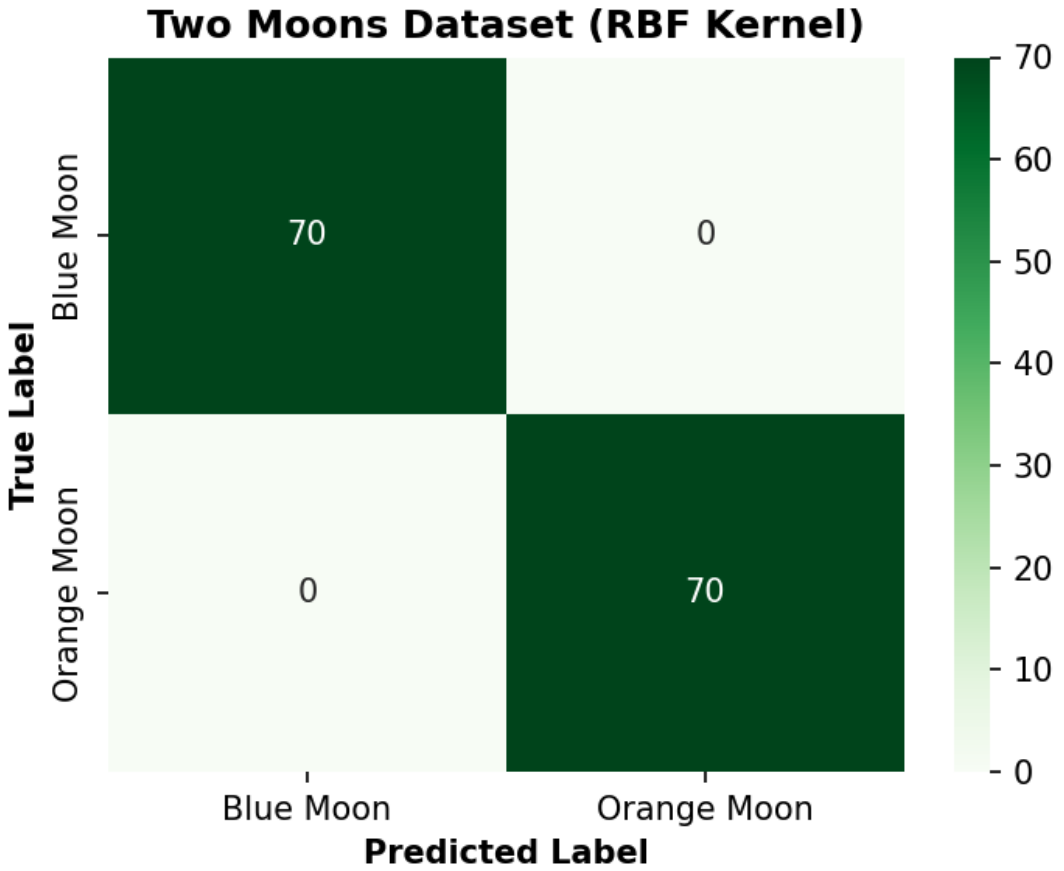
```
moon_rbf_preds = svm_rbf_moon.predict(Xm_test_scaled)
moon_rbf_preds
```

Output:

```
array([0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0,
0, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 1,
1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1,
0, 1, 1, 1, 1, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0,
1, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0,
1, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0,
1, 1, 1, 0, 1, 1, 0, 1])
```

## # Two Moons - RBF Kernel Confusion Matrix

```
class_names = ['Blue Moon', 'Orange Moon']

plt.figure(figsize=(5, 4))
sns.heatmap(
    confusion_matrix(ym_test, moon_rbf_preds),
    annot=True,
    fmt="d",
    cmap="Greens",
    xticklabels=class_names,
    yticklabels=class_names
)

plt.title("Two Moons Dataset (RBF Kernel)", fontweight='bold')
plt.xlabel("Predicted Label", fontweight='bold')
plt.ylabel("True Label", fontweight='bold')
plt.tight_layout()
plt.show()
```




**# Two Moons - RBF kernel classification metrics (Perfect accuracy!)**

```
print("Accuracy:", accuracy_score(ym_test, moon_rbf_preds))
print("Precision:", precision_score(ym_test, moon_rbf_preds, average="macro"))
print("Recall:", recall_score(ym_test, moon_rbf_preds, average="macro"))
print("F1 Score:", f1_score(ym_test, moon_rbf_preds, average="macro"))

print("\\nClassification Report:\\n")
print(classification_report(ym_test, moon_rbf_preds))
```

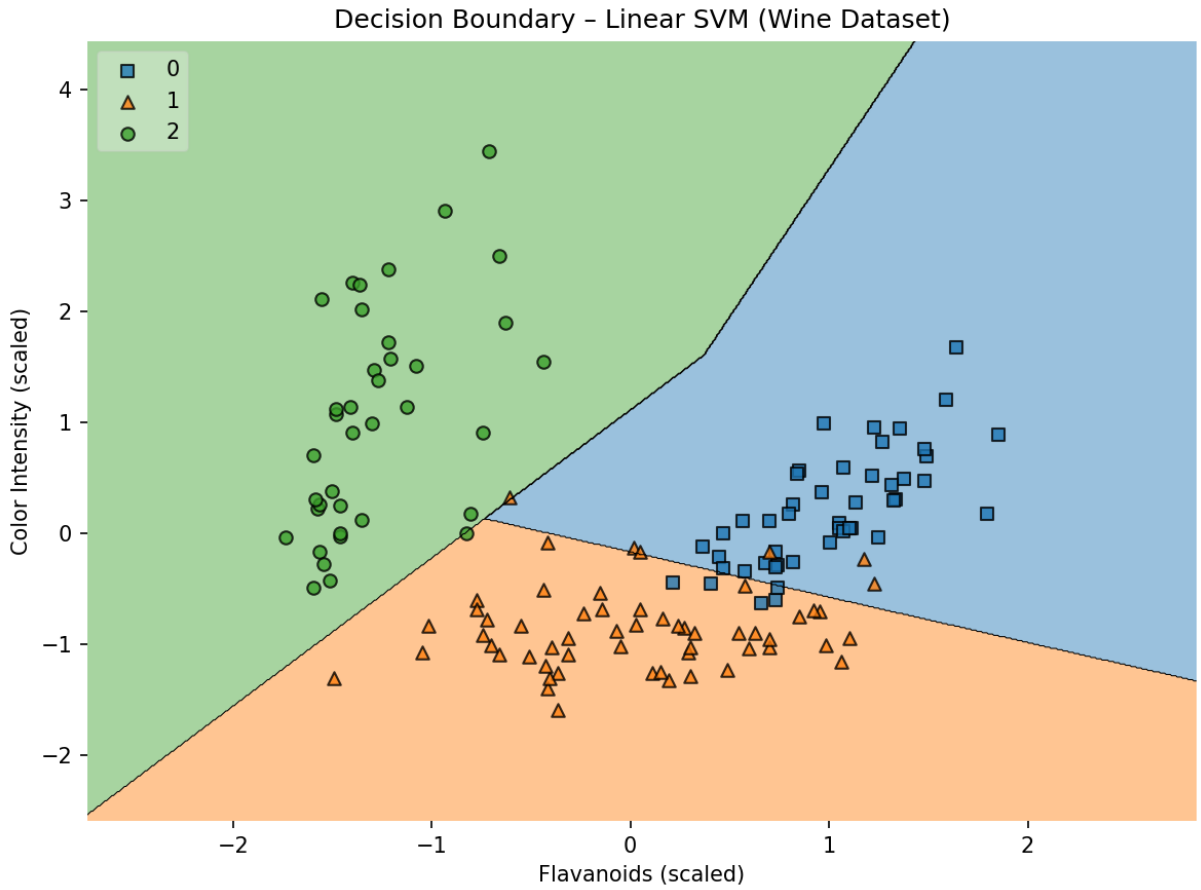
Output:

```
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0
Classification Report:
precision    recall  f1-score   support
0       1.00      1.00      1.00        70
1       1.00      1.00      1.00        70
accuracy                           1.00       140
macro avg       1.00      1.00      1.00       140
weighted avg       1.00      1.00      1.00       140
```

## Step 7: Model Simulation

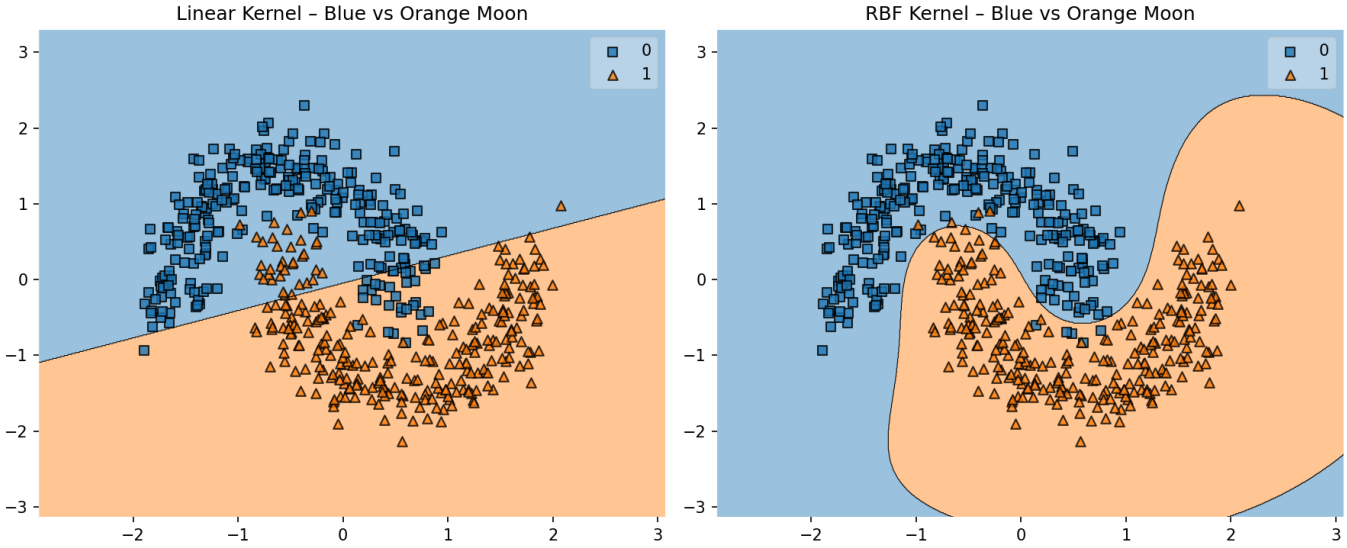
**# Model testing**

```
# Decision Boundaries - Wine
plot_decision_regions(
    Xw_train_scaled,
    yw_train.values,
    clf=svm_linear_wine,
    legend=2
)
plt.title("Decision Boundary - Linear SVM (Wine Dataset)")
plt.xlabel("Flavanoids (scaled)")
plt.ylabel("Color Intensity (scaled)")
plt.show()
```




**# Decision Boundaries - Two Moons**

```
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plot_decision_regions(Xm_train_scaled, ym_train, clf=svm_linear_moon)
plt.title("Linear Kernel - Blue vs Orange Moon")

plt.subplot(1,2,2)
plot_decision_regions(Xm_train_scaled, ym_train, clf=svm_rbf_moon)
plt.title("RBF Kernel - Blue vs Orange Moon")

plt.show()
```


Linear Kernel – Blue vs Orange Moon
RBF Kernel – Blue vs Orange Moon
0
1
3
2
1
0
−1
−2
−3
−2
−1
0
1
2
3